

UNNAM KARTHIKEYA

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PROFESSIONAL SUMMARY

Final-year CSE undergrad at IIIT Dharwad specializing in **GenAI engineering** and **cybersecurity** — building production-grade multi-agent LangGraph systems, hybrid RAG pipelines, and state-driven AI execution systems with measurable, benchmarked outcomes. Uniquely positioned at the intersection of secure AI development and full-stack deployment, with hands-on delivery of LLM-powered systems validated against real open-source codebases at sub-\$0.40 model cost per run.

PROJECTS

PatchForge AI — Autonomous Multi-Agent Code Repair *LangGraph · LiteLLM · Docker · PyGithub · Pydantic · LangSmith · SQLite*

- Architected** a 9-node stateful LangGraph pipeline (Triage → Research → Root-Cause → Planning → Patch → Repair Critic → Validation → PR) autonomously resolving GitHub issues — **1,436 passing tests, 0 retries**, Ruff/Mypy clean on pallets/click#3277 at **\$0.35 model cost**
- Eliminated** patch-application failures by replacing unified-diff patching with a pinned-SHA search-replace strategy grounded in verbatim file content fetched at `base_commit_sha` — **reduced patch errors to zero** across all benchmark runs
- Implemented** LangSmith observability with node-level latency telemetry, token accounting, and retry lineage; added SQLite repair memory storing successful strategies and failed patterns for future triage

QueryForge AI — Production-Grade Explainable RAG System *FastAPI · React · Pinecone · GPT-4o · LlamaIndex · LangChain · BM25*

- Built** a hybrid RAG pipeline combining Pinecone dense vector search with BM25 keyword retrieval — integrated parent/child context expansion, query classification (factual / summarization / citation-heavy), and adaptive `top_k` routing per query type
- Engineered** real-time SSE streaming via `POST /api/query/stream` delivering live GPT-4o tokens to React UI; implemented prompt injection defense sanitizing malicious instruction-like text in retrieved chunks before LLM generation
- Integrated** a continuous RAG evaluation pipeline tracking faithfulness, hallucination rate, groundedness, chunk recall, and p95 latency — surfaced in a React evaluation dashboard with benchmark trend charts

BOWA — AI-Powered Decision Intelligence & Execution System *FastAPI · LangChain · ChromaDB · Groq · PostgreSQL · WebSocket · Docker*

- Architected** a state-driven execution system across **38 service modules and 8,500+ lines** enforcing deliberate LLM boundaries — all intent detection, action selection, goal tracking, and state transitions fully deterministic; LLM invoked only at final response-rendering to eliminate hallucination-driven decision making
- Engineered** a 12-stage intelligence pipeline (Intent → State Evaluation → Reflection → Strategy → Plan → Action → Trajectory → LLM Render → Personality) with ChromaDB semantic memory and streak/reward behavioral feedback loops
- Delivered 151 automated tests** across 9 test files with full store isolation; built a background scheduler (5-min loop) with proactive notification engine, watch-to-action auto-conversion, and adaptive news intelligence pipeline

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, HTML5, CSS3
AI / GenAI: LangGraph, LangChain, LlamaIndex, LiteLLM, OpenAI API, Anthropic Claude API, RAG, Hybrid Retrieval (BM25 + Dense), Prompt Engineering, Vector Databases (Pinecone, ChromaDB, FAISS), Multi-Agent Systems, Agentic Workflows, SSE Streaming
Frameworks & Tools: FastAPI, React, Node.js, Express.js, Docker, Pydantic, PostgreSQL, SQLite, PyGithub, Groq, Ollama, Tavily, LangSmith, WebSocket, Ruff, Mypy, pytest, Git
Cybersecurity: Malware Analysis, YARA Rules, MITRE ATT&CK, Threat Intelligence, Static & Dynamic Analysis, x64dbg, Wireshark

EDUCATION

Indian Institute of Information Technology, Dharwad B.Tech, Computer Science and Engineering	2023 – 2027 Expected May 2027
Narayana Saraswati Bhavan, Vijayawada Intermediate — MPC	2021 – 2023
Gowtham Model School Secondary School Certificate (SSC)	Passed 2021